Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Designing Mechanically Intelligent Robotic Manipulators for Highly Unstructured Environments			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robot Design Technologies	Robotics/Automation Machinery	Robot Control and Intelligence Technologies
Concept of Proposed Technology	Mechanical intelligence is the concept of developing new robotic platforms and mechanisms that make challenges for perception, control and autonomy easier or more robust for natural, unstructured, and often unpredictable environments by the virtue of their physical designs. Examples of such environments can range from space exploration, manufacturing, to typical household. These mechanically intelligent hardware can solve unique grasping challenges in an inherently robust and efficient way, alleviating the limitations of traditional, off-the-shelf hardware and working synergistically well with state-of-the-art AI algorithms for perceptions and controls. Below are some research topics for designing and building mechanically intelligence manipulators for highly unstructured environments, which can be at home, or in the field: • **Soft universal manipulator for robots at home:** Universal grippers are extremely			

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	versatile at grasping various objects, which is uniquely suited for domestic robots. However, they currently lack any in-hand manipulation capabilities, which makes important tasks such as opening a medicine bottle difficult. These soft universal manipulators are also inherently soft, which makes them safer to interact daily with humans. ● **Rolling contact joints for grasping:** Like human joints, rolling contact joints rely on rolling contact between two surfaces secured by tendons. This joint is currently under-explored in robotics, but it offers many unique advantages for manipulation, such as ultra-low friction, inherent robustness, and safety. In addition, there is untapped potential in incorporating non-circular rolling contact joints, which can mechanically encode force-torque transmission ratios along a single trajectory. ● **Equipping different locomotion platforms with manipulation capability:** Highly unstructured environments often demand unique manipulation solutions. An active research area in the lab is to equip different locomotion platforms (such as drones) with manipulation capability, so they can perform tasks beyond their traditional capability (such as aerial grasping).
Rationale for Joint R&D Project	The proposed joint R&D projects greatly benefit from the close collaboration between the research institute and the Korean SME partner, because it grounds the research in real-world applications that can be immediately utilized by the Korean SME partner. This is important for the research institution as well because it ensures that research is aimed at the areas of robotics that can provide the most benefits in the short term. This mutually beneficial collaboration will speed up the process of implementing cutting-edge robotics hardware research into deployable robotic prototypes and systems.
Key Words	1. Mechanical Intelligence 2. Locomotion 3. Grasping and Manipulation 4. Robots in Complex and Extreme Environments 5. Physical AI

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]		**Start of the Project**	Level (2 -3)	**End of the Project**	Level (6)
R&D Objectives		The main goal of this R&D is to tackle the problem that most of the robotics community and research is still using off-the-shelf hardware to solve some of the really challenging grasping and manipulation problems, and are often limited by them. The specific R&D objectives are: ● Derive design specifications based on the particular grasping challenge and environment. ● Understand and model desired contact dynamics to achieve the design specifications. ● Design and prototype different gripper morphology to achieve robust performance. ● Investigate tactile sensing needs for AI-synergy.			
R&D Target Specifications		The target specifications for this R&D effort will be heavily focused on prototyping physical prototypes, and will vary based on the exact research application area. In general, to evaluate grasping and manipulation, the following criteria are used: (i) grasp quality and success rate, (ii) robustness to external disturbance, (iii) physical safety if interacting with other robots or humans, (iv) in-hand manipulation capability and speed.			
R&D Plan		This R&D research effort will be over the course of three years, following an iterative design approach through low-fidelity, rapid prototyping. Year 1 will be focused on gathering and understanding the specific design challenges and constraints provided by the Korean SME partner, understanding the fundamental physical principles, and modeling the contact dynamics and potential solution. Year 2 will focus on rapid prototyping and iterative design, and evaluation in a laboratory setting. Year 3 will focus on implementing the design solution in a real-world setting, evaluating and refining the design.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for developing the fundamental technology, studying the contact dynamics derived from the specifications and application scenarios provided by the Korean SME partner through modeling and simulation, and more importantly, design and prototype functional hardware that can be tested in a laboratory setting. The research institute is also responsible for transferring the fundamental findings of this research initiative to the Korean SME partner.
Expected R&R of Korean SME Partner	This is expected to be a close working relationship between the Korean SME partner and the Research Institute. This close collaboration will foster collective innovation and ensure that the developing technology fits the needs of the program. The Korean SME partner will be responsible for providing application details from which the research institute can derive design specifications . The Korean SME is also expected to provide potential engineering support (if needed) so that the research institute can ensure the research direction fits within the scope of the program, along with maximizing the research potential of the research institute.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcome of this joint research effort would be the delivery of unique hardware prototypes to solve some of the most challenging manipulation problems robots face today. These hardware prototypes will not only make control and perception challenges way simpler but also work synergistically well with state-of-the-art AI/ML algorithms to amplify the potential impact. This will allow the Korean SME partners to no longer only rely on off-the-shelf robotic hardware and traditional solutions and support rapid technology commercialization.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization